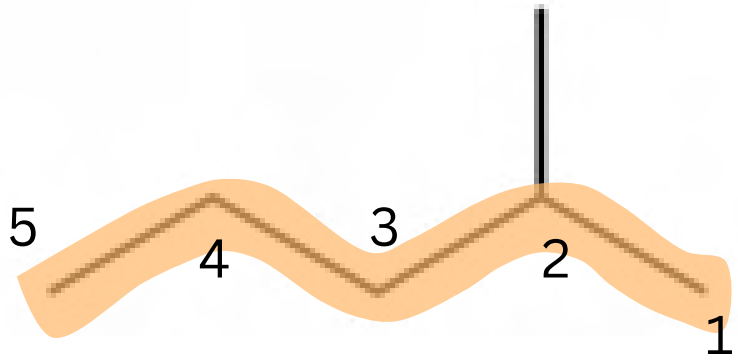
CARD A01 – Structure → Name

Level: Foundation Molecular Formula: C6H14

1. NUMBERED PARENT



2. NAMING TRAIL

Parent: longest chain = 5 C (pentane) and carries one methyl at 2C

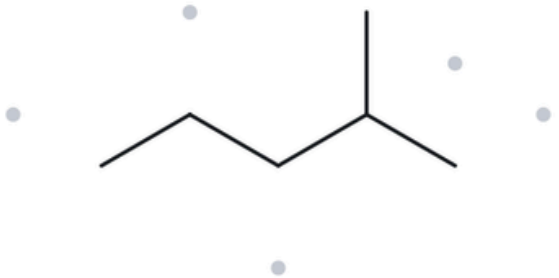
Numbering: right to left

Substituents: methyl on C2 (2-methyl)

3. MY NAME

2-methylpentane

4. THE ALKANE CHECK



2-methylpentane

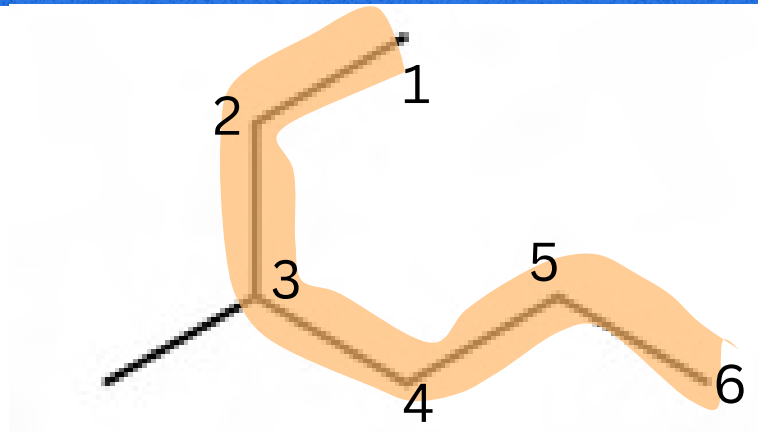
Match

✓ longest chain confirmed ✓ lowest locant ✓ lowest alphabetical order ✓ punctuation ✓ formula counted

CARD A02 – Structure → Name

Level: Foundation Molecular Formula: C7H16

1. NUMBERED PARENT



2. NAMING TRAIL

Parent: longest chain = 6 C (hexane) and carries one methyl at 3C

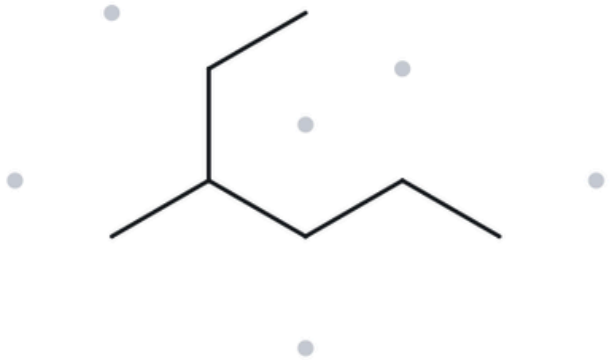
Numbering: top tip and go downward, then to the right

Substituent: methyl on C3 (3-methyl)

3. MY NAME

3-methylhexane

4. THE ALKANE CHECK



3-methylhexane

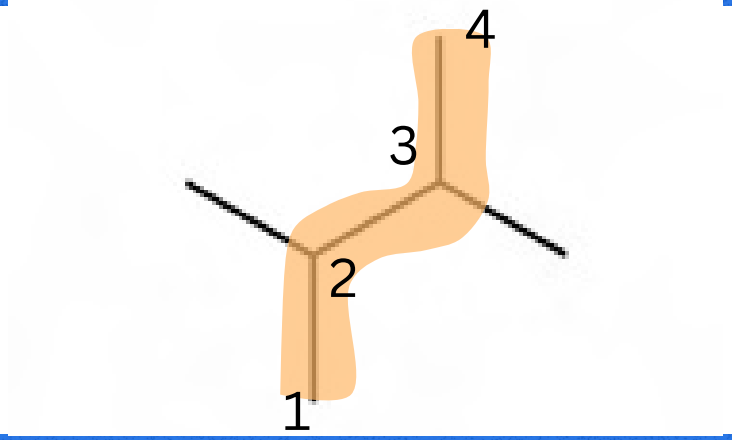
Match

✓ longest chain confirmed ✓ lowest locant ✓ lowest alphabetical order ✓ punctuation ✓ formula counted

CARD A03 – Structure → Name

Level: Foundation Molecular Formula: C6H14

1. NUMBERED PARENT



2. NAMING TRAIL

Parent: longest chain = 4 C (butane) and carries two methyl

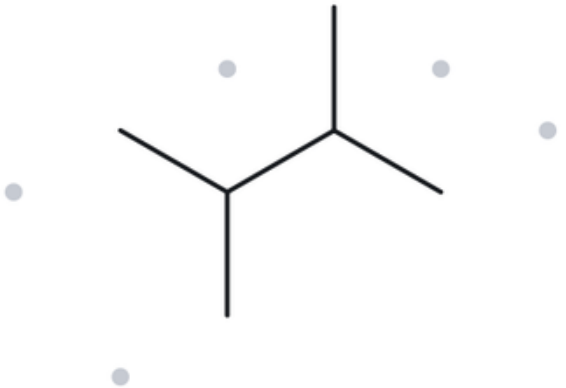
Numbering: top tip then go downward or vice versa

Substituent: methyl groups on 2C and 3C (2,3-dimethyl)

3. MY NAME

2,3-dimethylbutane

4. THE ALKANE CHECK



2,3-dimethylbutane

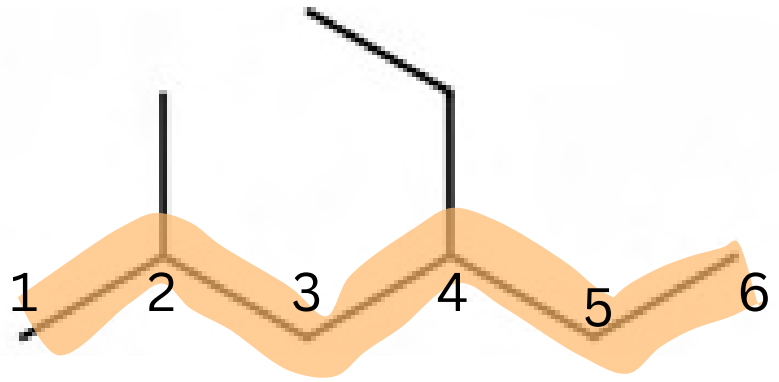
Match

✓ longest chain confirmed ✓ lowest locant ✓ lowest alphabetical order ✓ punctuation ✓ formula counted

CARD A04 – Structure → Name

Level: Developing Molecular Formula: C9H20

1. NUMBERED PARENT



2. NAMING TRAIL

Parent: longest chain = 6 C (hexane) and carries one methyl at 2C and one ethyl at 4C

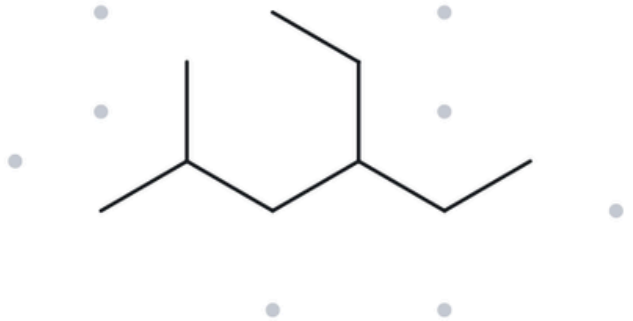
Numbering: left to right

Substituent: one methyl on 2C and one ethyl on 4C (4-ethyl-2-methyl)

3. MY NAME

4-ethyl-2-methylhexane

4. THE ALKANE CHECK



4-ethyl-2-methylhexane

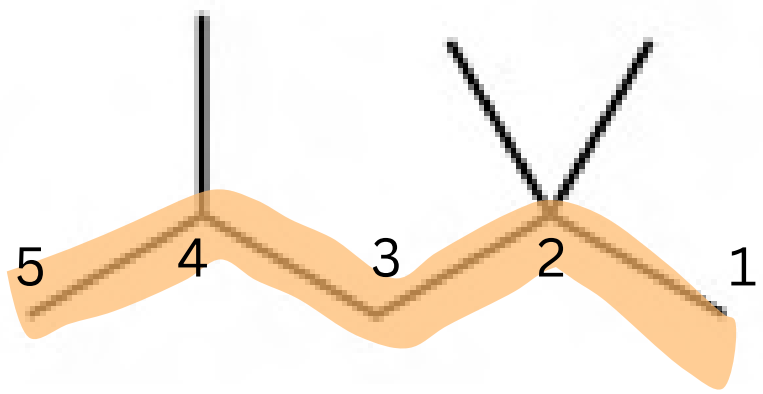
Match

✓ longest chain confirmed ✓ lowest locant ✓ lowest alphabetical order ✓ punctuation ✓ formula counted

CARD A05 – Structure → Name

Level: Developing Molecular Formula: C8H18

1. NUMBERED PARENT



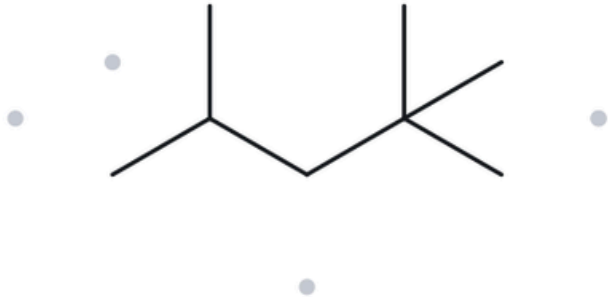
2. NAMING TRAIL

Parent: longest chain = 5C (pentane) and carries two methyl on 2C and another one on 4C
Numbering: right to left or vice versa
Substituent: 2,2,4-trimethyl

3. MY NAME

2,2,4-trimethylpentane

4. THE ALKANE CHECK



2,2,4-trimethylpentane

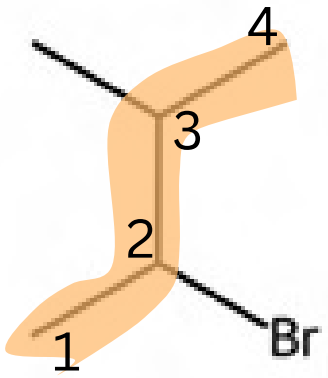
Match

✓ longest chain confirmed ✓ lowest locant ✓ lowest alphabetical order ✓ punctuation ✓ formula counted

CARD A06 – Structure → Name

Level: Developing Molecular Formula: C5H11Br

1. NUMBERED PARENT



2. NAMING TRAIL

Parent: longest chain = 4 C (butane)

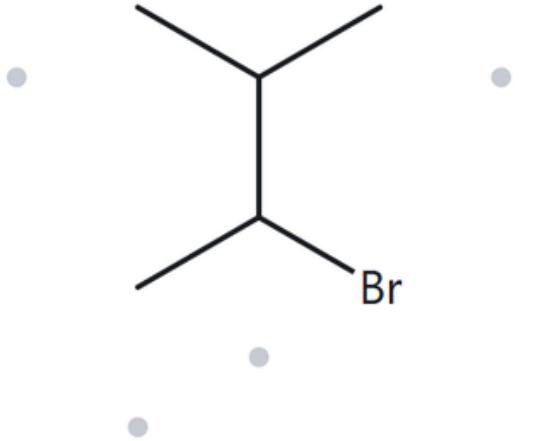
Numbering: top right to bottom left or vice versa

Substituent: bromo (halogen) on 2C and methyl on 3C (2-bromo-3-methylbutane)

3. MY NAME

2-bromo-3-methylbutane

4. THE ALKANE CHECK



2-bromo-3-methylbutane

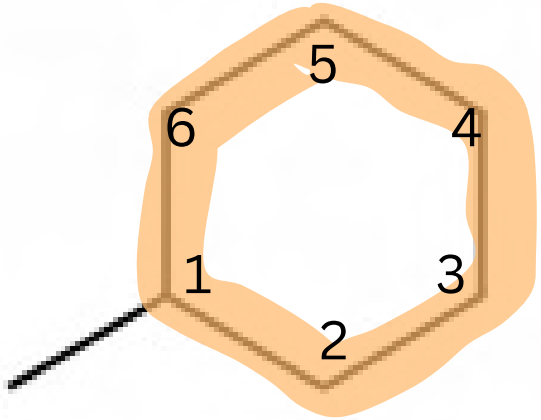
Match

✓ longest chain confirmed ✓ lowest locant ✓ lowest alphabetical order ✓ punctuation ✓ formula counted

CARD A07 – Structure → Name

Level: Developing Molecular Formula: C7H14

1. NUMBERED PARENT



2. NAMING TRAIL

Parent: 6C ring (cyclohexane) and carries one methyl.

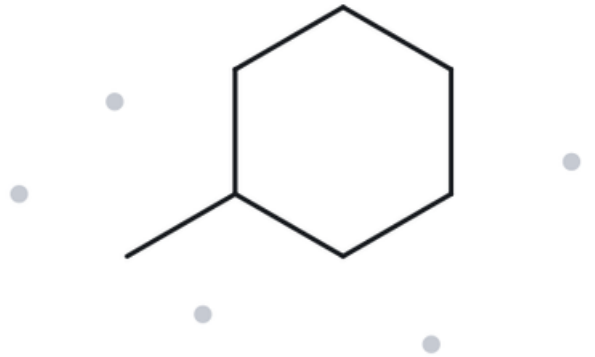
Numbering: around the ring, starting at the substituent carbon, then go in either direction

Substituent: methyl on 1C (methyl)

3. MY NAME

methylcyclohexane

4. THE ALKANE CHECK



methylcyclohexane

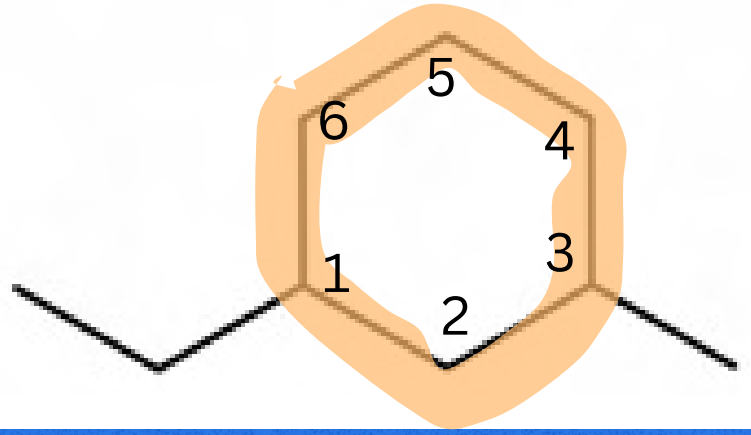
Match

✓ longest chain confirmed ✓ lowest locant ✓ lowest alphabetical order ✓ punctuation ✓ formula counted

CARD A08 – Structure → Name

Level: Challenge Molecular Formula: C9H18

1. NUMBERED PARENT



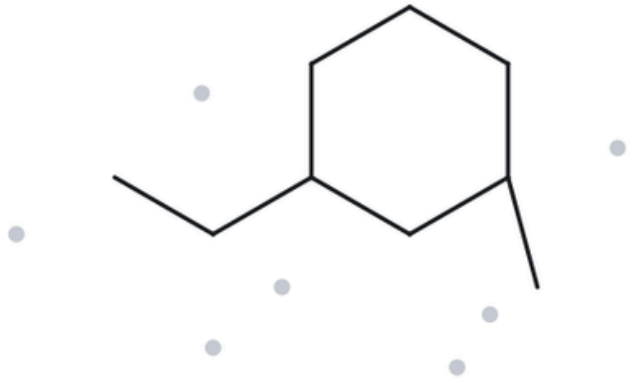
2. NAMING TRAIL

Parent: longest chain = 6C ring (cyclohexane)
Numbering: start at ethyl and go to methyl (path with shortest distance between them) then go around the ring
Substituent: ethyl on 1C and methyl on 3C (1-ethyl-3-methyl)

3. MY NAME

1-ethyl-3-methylcyclohexane

4. THE ALKANE CHECK



1-ethyl-3-methylcyclohexane

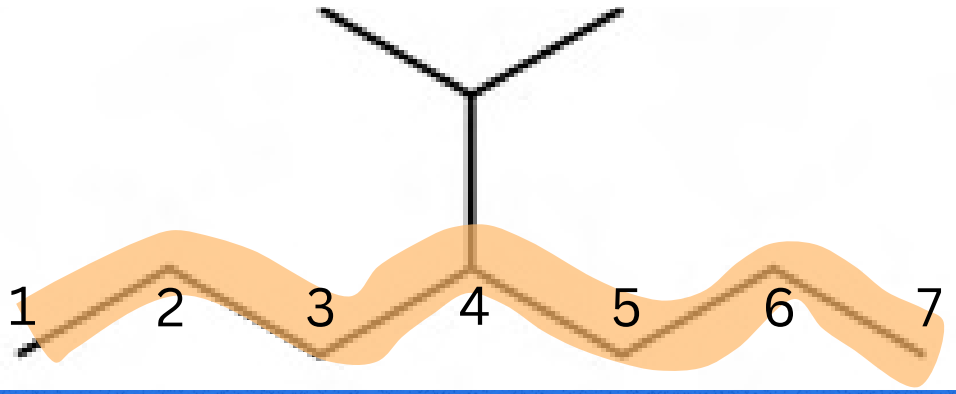
Match

✓ longest chain confirmed ✓ lowest locant ✓ lowest alphabetical order ✓ punctuation ✓ formula counted

CARD A09 – Structure → Name

Level: Challenge Molecular Formula: C10H22

1. NUMBERED PARENT



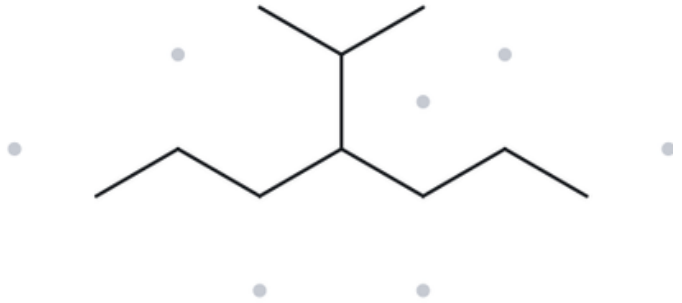
2. NAMING TRAIL

Parent: longest chain = 7C (heptane)
Numbering: left to right or vice versa
Substituent: isopropyl on 4C (4-isopropyl)

3. MY NAME

4-isopropylheptane

4. THE ALKANE CHECK



✗ Mismatch. The tool chose a 6-carbon parent (hexane), but the longest chain is 7 carbons (heptane). Correct name: **4-isopropylheptane**.

2-methyl-3-propylhexane

✓ longest chain confirmed ✓ lowest locant ✓ lowest alphabetical order ✓ punctuation ✓ formula counted

CARD A10 – Structure → Name

Level: Challenge Molecular Formula: C7H14

1. NUMBERED PARENT



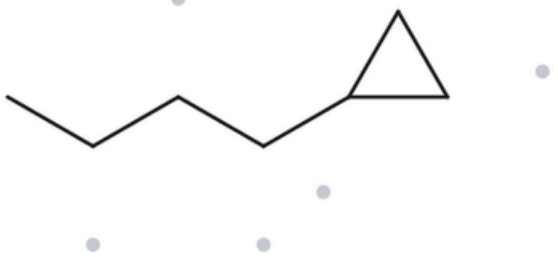
2. NAMING TRAIL

Parent: 3C ring (cyclopropane) is the parent even though the butyl chain (4C) is longer
Numbering: start from substituent carbon then go either way
Substituent: butyl at C1 (butyl)

3. MY NAME

butylcyclopropane

4. THE ALKANE CHECK



butylcyclopropane

Match

✓ longest chain confirmed ✓ lowest locant ✓ lowest alphabetical order ✓ punctuation ✓ formula counted